

**Crowding baseline**  $\mu_k(x_k(t))$

tier imbalance, concentration,  
leverage, cross-asset spillover

*CovTPP / ST-GNN / LGBM add this term*

**Self-excitation**

$\sum_{k'} \sum_{t_j < t} \alpha_{k' \rightarrow k} \phi(t - t_j)$   
exponential kernel over past events

*Hawkes / THP capture this term only*

$\lambda_k(t)$

conditional burst intensity